

Limit

Exercise 17.1

1. Evaluate

$$\begin{array}{ll} \textcircled{1} \text{ a) } \lim_{n \rightarrow 2} (4n^2 - 3n + 5) & \textcircled{2} \text{ b) } \lim_{n \rightarrow 0} (9n^2 - 5n + 1) \\ = 4 \times 2^2 - 3 \times 2 + 5 & = 9 \times 0^2 - 5 \times 0 + 1 \\ = 16 - 6 + 5 & = 0 - 0 + 1 \\ = 16 - 1 & = 1 \\ = 15 & \end{array}$$

$$\begin{array}{l} \textcircled{3} \text{ c) } \lim_{n \rightarrow 1} \frac{9n - 7}{4n + 3} \\ = \frac{9 \times 1 - 7}{4 \times 1 + 3} \\ = \frac{9 - 7}{4 + 3} \\ = \frac{2}{7} \end{array}$$

2. Evaluate the following limit

$$\textcircled{4} \text{ a) } \lim_{n \rightarrow 0} \frac{5n^2 - 3n}{9n}$$

→ The function takes $\frac{0}{0}$ form (indeterminate) when $n=0$

$$= \lim_{n \rightarrow 0} \frac{n(5n - 3)}{9n}$$

$$= \lim_{n \rightarrow 0} \frac{5n-3}{9}$$

$$= \frac{5 \times 0 - 3}{9}$$

$$= \frac{-3}{9}$$

$$= -\frac{1}{3} \quad \text{Ans.}$$

5) b) $\lim_{n \rightarrow 1} \frac{n^2 - 5n + 4}{n^2 + 2n - 3}$

→ The function takes $\frac{0}{0}$ form (indeterminate)

When $n=1$

$$= \lim_{n \rightarrow 1} \frac{n^2 - 4n - n + 4}{n^2 + 3n - n - 3}$$

$$= \lim_{n \rightarrow 1} \frac{n(n-4) - 1(n-4)}{n(n+3) - 1(n+3)}$$

$$= \lim_{n \rightarrow 1} \frac{\cancel{n-1} (n-4)}{\cancel{n-1} (n+3)}$$

$$= \frac{1-4}{1+3}$$

$$= -\frac{3}{4} \quad \text{Ans.}$$

6) $\lim_{n \rightarrow 1} \frac{n^3 - 1}{n - 1}$

→ The function takes $\frac{0}{0}$ form (indeterminate) when $n=1$

$$= \lim_{n \rightarrow 1} \frac{(n-1)(n^2+2n+1)}{(n-1)}$$

$$= 1^2 + 2 \times 1 + 1$$

$$= 1 + 2 + 1$$

$$= 4$$

7) $\lim_{n \rightarrow a} \frac{n^{2/3} - a^{2/3}}{n - a}$

→ The function takes $\frac{0}{0}$ form (indeterminate) when $n=a$

$$= \lim_{n \rightarrow a} \frac{n^{2/3} - a^{2/3}}{n - a}$$

$$= \frac{2}{3} a^{2/3-1}$$

$$= \frac{2}{3} a^{-1/3}$$

$$= \frac{2}{3a^{1/3}}$$



8 e) $\lim_{n \rightarrow a} \frac{n^5 - a^5}{n^4 - a^4}$

→ The function takes $\frac{0}{0}$ form (indetermin-

ate) when $n=a$

$$= \lim_{n \rightarrow a} \frac{n^5 - a^5}{\frac{n^4 - a^4}{n - a}}$$

$$= \lim_{n \rightarrow a} \frac{5a^{5-1}}{4a^{4-1}} \left(\text{use } \lim_{n \rightarrow a} \frac{n^n - a^n}{n - a} = na^{n-1} \right)$$

$$= \frac{5a^4}{4a^3}$$

$$= \frac{5a}{4} \quad \text{Ans.}$$

9 f) $\lim_{n \rightarrow 0} \frac{(n+1)^8 - 1}{(n+1)^2 - 1}$

→ The function takes $\frac{0}{0}$ form (indetermin-

ate) when $n=0$

$$= \lim_{n \rightarrow 0} \frac{(n+1)^8 - 1^8}{(n+1)^2 - 1^2} \quad [\because 1^8 = 1^2 = 1]$$

$$= \lim_{n \rightarrow 0} \frac{\frac{(n+1)^8 - 1^8}{(n+1) - 1}}{\frac{(n+1)^2 - 1^2}{(n+1) - 1}}$$

$$= \lim_{x+1 \rightarrow 0+1} \frac{(x+1)^8 - 1^8}{\frac{(x+1) - 1}{(x+1)^2 - 1^2} \cdot (x+1) - 1}$$

$$= \frac{8 \cdot 1^7}{2 \cdot 1^1} \left(\text{using } \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right)$$

$$= \frac{8 \cdot 1}{2}$$

$$= 4$$

eg) $\lim_{x \rightarrow 64} \frac{\sqrt[6]{x} - 2}{\sqrt[3]{x} - 4}$

→ The function takes $\frac{0}{0}$ form (indeterminate) when $x = 64$

$$= \lim_{x \rightarrow 64} \frac{x^{1/6} - 2}{x^{1/3} - 4}$$

$$= \lim_{x \rightarrow 64} \frac{x^{1/6} - 64^{1/6}}{x^{1/3} - 64^{1/3}}$$

$$= \lim_{x \rightarrow 64} \frac{x^{1/6} - 64^{1/6}}{\frac{x - 64}{x^{1/3} - 64^{1/3}} \cdot (x - 64)}$$

$$= \frac{\frac{1}{6} \times 64^{1/6-1}}{\frac{1}{3} \times 64^{1/3-1}} \left(\text{using } \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right)$$

$$= \frac{\frac{1}{82} \times 8 \times 64^{-5/6}}{64^{-2/3}}$$

$$= \frac{1}{2} \times 64^{2/3 - 5/6}$$

$$= \frac{1}{2} \times 64^{-1/6}$$

$$= \frac{1}{2} \times 2^{8 \times -1/6}$$

$$= \frac{1}{2} \times \frac{1}{2}$$

$$= \frac{1}{4} \text{ \#.}$$

iii) $\lim_{n \rightarrow 3} \left(\frac{1}{n-3} - \frac{9}{n^3-3n^2} \right)$

→ ~~The~~ The function take infine form (∞) when $n=3$.

$$= \lim_{n \rightarrow 3} \left(\frac{1}{n-3} - \frac{9}{n^2(n-3)} \right)$$

$$= \lim_{n \rightarrow 3} \left(\frac{n^2-9}{n^2(n-3)} \right)$$

$$= \lim_{n \rightarrow 3} \left(\frac{(n+3)(n-3)}{n^2(n-3)} \right)$$

$$= \lim_{n \rightarrow 3} \left(\frac{n+3}{n^2} \right)$$

$$= \frac{3+3}{3^2}$$

$$= \frac{\cancel{6} \cdot 2}{\cancel{9} \cdot 3}$$

$$= \frac{2}{3} \neq$$

3. Evaluate the following

12 a) $\lim_{n \rightarrow 25} \frac{n-25}{\sqrt{n}-5}$

→ The function takes indeterminate form when $n=25$

$$= \lim_{n \rightarrow 25} \frac{(\sqrt{n})^2 - 5^2}{\sqrt{n} - 5}$$

$$= \lim_{n \rightarrow 25} \frac{(\sqrt{n} + 5)(\sqrt{n} - 5)}{(\sqrt{n} - 5)}$$

$$= \sqrt{25} + 5$$

$$= 5 + 5$$

$$= 10 \neq$$

13 b) $\lim_{n \rightarrow 0} \frac{\sqrt{1+2n} - \sqrt{1-3n}}{n}$

→ The function takes indeterminate form when $n=0$

$$= \lim_{n \rightarrow 0} \frac{\sqrt{1+2n} - \sqrt{1-3n}}{n} \times \frac{\sqrt{1+2n} + \sqrt{1-3n}}{\sqrt{1+2n} + \sqrt{1-3n}}$$

$$= \lim_{n \rightarrow 0} \frac{(\sqrt{1+2n})^2 - (\sqrt{1-3n})^2}{n[\sqrt{1+2n} + \sqrt{1-3n}]}$$

$$= \lim_{x \rightarrow 0} \frac{x+2x - x+3x}{x(\sqrt{1+2x} + \sqrt{1-3x})}$$

$$= \lim_{x \rightarrow 0} \frac{5x}{x(\sqrt{1+2x} + \sqrt{1-3x})}$$

$$= \frac{5}{\sqrt{1+2 \cdot 0} + \sqrt{1-3 \cdot 0}}$$

$$= \frac{5}{1+1}$$

$$= \frac{5}{2}$$

④ c) $\lim_{x \rightarrow 1} \frac{\sqrt{3x+1} - 2}{x-1}$

→ The function takes indeterminate form when $x=1$

$$= \lim_{x \rightarrow 1} \frac{\sqrt{3x+1} - 2}{x-1} \times \frac{\sqrt{3x+1} + 2}{\sqrt{3x+1} + 2}$$

$$= \lim_{x \rightarrow 1} \frac{(\sqrt{3x+1})^2 - 2^2}{(x-1)(\sqrt{3x+1} + 2)}$$

$$= \lim_{x \rightarrow 1} \frac{3x+1-4}{(x-1)(\sqrt{3x+1} + 2)}$$

$$= \lim_{x \rightarrow 1} \frac{3x-3}{(x-1)(\sqrt{3x+1} + 2)}$$

$$= \lim_{x \rightarrow 1} \frac{3(x-1)}{(x-1)(\sqrt{3x+1} + 2)}$$

$$= \frac{3}{\sqrt{3 \times 1 + 1} + 2}$$

$$= \frac{3}{2+2}$$

$$= \frac{3}{4} \quad \#.$$

⑤ d) $\lim_{n \rightarrow a} \frac{\sqrt{n+a} - \sqrt{3n-a}}{n-a}$

→ The function takes indeterminate form when $n=a$

$$= \lim_{n \rightarrow a} \frac{\sqrt{n+a} - \sqrt{3n-a}}{n-a} \times \frac{\sqrt{n+a} + \sqrt{3n-a}}{\sqrt{n+a} + \sqrt{3n-a}}$$

$$= \lim_{n \rightarrow a} \frac{(\sqrt{n+a})^2 - (\sqrt{3n-a})^2}{(n-a)(\sqrt{n+a} + \sqrt{3n-a})}$$

$$= \lim_{n \rightarrow a} \frac{n+a - 3n+a}{(n-a)(\sqrt{n+a} + \sqrt{3n-a})}$$

$$= \lim_{n \rightarrow a} \frac{-2(n-a)}{(n-a)(\sqrt{n+a} + \sqrt{3n-a})}$$

$$= \frac{-2}{\sqrt{a+a} + \sqrt{3a-a}}$$

$$= \frac{-2}{2\sqrt{2a}}$$

$$= \frac{-1}{\sqrt{2a}} \quad \#.$$

⑥ e) $\lim_{x \rightarrow 2} \frac{x^2 - 4}{\sqrt{x+2} - \sqrt{3x-2}}$

→ The function takes indeterminate form when $x=2$

$$= \lim_{x \rightarrow 2} \frac{x^2 - 4}{\sqrt{x+2} - \sqrt{3x-2}} \times \frac{\sqrt{x+2} + \sqrt{3x-2}}{\sqrt{x+2} + \sqrt{3x-2}}$$

$$= \lim_{x \rightarrow 2} \frac{(x^2 - 4)(\sqrt{x+2} + \sqrt{3x-2})}{(\sqrt{x+2})^2 - (\sqrt{3x-2})^2}$$

$$= \lim_{x \rightarrow 2} \frac{(x^2 - 4)(\sqrt{x+2} + \sqrt{3x-2})}{x+2 - 3x+2}$$

$$= \lim_{x \rightarrow 2} \frac{(x+2)(x-2)(\sqrt{x+2} + \sqrt{3x-2})}{-2x - 2(x-2)}$$

$$= \frac{(2+2)(\sqrt{2+2} + \sqrt{3 \times 2 - 2})}{-2}$$

$$= \frac{4^2(\sqrt{4} + \sqrt{4})}{-2}$$

$$= -2(2+2)$$

$$= -2 \times 4$$

$$= -8$$

$$\textcircled{15} f) \lim_{x \rightarrow 1} \frac{\sqrt{x^2+15} - 4}{x-1}$$

→ The function takes indeterminate form when $x=1$

$$= \lim_{x \rightarrow 1} \frac{\sqrt{x^2+15} - 4}{x-1} \times \frac{\sqrt{x^2+15} + 4}{\sqrt{x^2+15} + 4}$$

$$= \lim_{x \rightarrow 1} \frac{(\sqrt{x^2+15})^2 - 4^2}{(x-1)(\sqrt{x^2+15} + 4)}$$

$$= \lim_{x \rightarrow 1} \frac{x^2+15-16}{(x-1)(\sqrt{x^2+15} + 4)}$$

$$= \lim_{x \rightarrow 1} \frac{x^2-1}{(x-1)(\sqrt{x^2+15} + 4)}$$

$$= \lim_{x \rightarrow 1} \frac{(x+1)(x-1)}{(x-1)(\sqrt{x^2+15} + 4)}$$

$$= \frac{1+1}{\sqrt{1^2+15} + 4}$$

$$= \frac{2}{\sqrt{16} + 4}$$

$$= \frac{2}{4+4}$$

$$= \frac{2}{8}$$

$$= \frac{1}{4} \#.$$

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$$(18) g) \lim_{x \rightarrow 0} \frac{\sqrt{3+x^2} - \sqrt{3-x^2}}{x}$$

→ The function takes indeterminate form when $x=0$

$$= \lim_{x \rightarrow 0} \frac{\sqrt{3+x^2} - \sqrt{3-x^2}}{x} \times \frac{\sqrt{3+x^2} + \sqrt{3-x^2}}{\sqrt{3+x^2} + \sqrt{3-x^2}}$$

$$= \lim_{x \rightarrow 0} \frac{(\sqrt{3+x^2})^2 - (\sqrt{3-x^2})^2}{x(\sqrt{3+x^2} + \sqrt{3-x^2})}$$

$$= \lim_{x \rightarrow 0} \frac{x^2 + x^2 - x^2 + x^2}{x(\sqrt{3+x^2} + \sqrt{3-x^2})}$$

$$= \lim_{x \rightarrow 0} \frac{2x^2}{x(\sqrt{3+x^2} + \sqrt{3-x^2})}$$

$$= \frac{2 \times 0}{(\sqrt{3 \times 0} + \sqrt{3-0})}$$

$$= 0$$

$$(19) h) \lim_{x \rightarrow 0} \frac{\sqrt{x^2+4} - 2}{x^2}$$

→ The function takes indeterminate form when $x=0$

$$= \lim_{x \rightarrow 0} \frac{\sqrt{x^2+4} - 2}{x^2} \times \frac{\sqrt{x^2+4} + 2}{\sqrt{x^2+4} + 2}$$

$$= \lim_{x \rightarrow 0} \frac{(\sqrt{x^2+4})^2 - 2^2}{x^2(\sqrt{x^2+4} + 2)}$$

$$= \lim_{x \rightarrow 0} \frac{x^2 + 4 - 4}{x^2(\sqrt{x^2+4} + 2)}$$

$$= \lim_{x \rightarrow 0} \frac{x^2}{x^2(\sqrt{x^2+4} + 2)}$$

$$= \frac{1}{\sqrt{0^2+4} + 2}$$

$$= \frac{1}{2+2}$$

$$= \frac{1}{4} \text{ Ans.}$$

20 i) $\lim_{x \rightarrow 1} \frac{x - \sqrt{2-x^2}}{2x - \sqrt{2+2x^2}}$

→ The function takes indeterminate form when $x=1$

$$= \lim_{x \rightarrow 1} \frac{x - \sqrt{2-x^2}}{2x - \sqrt{2+2x^2}} \times \frac{x + \sqrt{2-x^2}}{x + \sqrt{2-x^2}} \times \frac{2x + \sqrt{2+2x^2}}{2x + \sqrt{2+2x^2}}$$

$$= \lim_{x \rightarrow 1} \frac{x^2 - (\sqrt{2-x^2})^2}{(2x)^2 - (\sqrt{2+2x^2})^2} \times \frac{2x + \sqrt{2+2x^2}}{x + \sqrt{2-x^2}}$$

$$= \lim_{x \rightarrow 1} \frac{x^2 - 2 + x^2}{4x^2 - 2 - 2x^2} \times \frac{2x + \sqrt{2+2x^2}}{x + \sqrt{2-x^2}}$$

$$= \lim_{x \rightarrow 1} \frac{2x^2 - 2}{2x^2 - 2} \times \frac{2x + \sqrt{2+2x^2}}{x + \sqrt{2-x^2}}$$

$$= \frac{2 \times 1 + \sqrt{2+2 \times 1^2}}{1 + \sqrt{2-1^2}}$$

$$= \frac{2+2}{1+1}$$

$$= \frac{4}{2}$$

$$= 2 \text{ Ans.}$$

4. Evaluating the following limits.

21 a) $\lim_{n \rightarrow \infty} \frac{5n^2}{3n^2 + 9}$

→ The function takes $\frac{\infty}{\infty}$ form (indeterminate)

$$= \lim_{n \rightarrow \infty} \frac{5n^2}{n^2 \left(3 + \frac{9}{n^2} \right)}$$

$$= \frac{5}{\frac{3+9}{\infty}}$$

$$= \frac{5}{3+0}$$

$$= \frac{5}{3} \quad \#.$$

22 b) $\lim_{n \rightarrow \infty} \frac{\sqrt{n^2 + 1}}{n + 1}$

→ The function takes $\frac{\infty}{\infty}$ form (indeterminate)

$$= \lim_{n \rightarrow \infty} \frac{\sqrt{n^2 \left(1 + \frac{1}{n^2} \right)}}{n + 1}$$

$$= \lim_{n \rightarrow \infty} \frac{n \sqrt{\left(1 + \frac{1}{n^2} \right)}}{n \left(1 + \frac{1}{n} \right)}$$

$$= \frac{1 + \frac{1}{\infty}}{1 + \frac{1}{\infty}}$$

$$= \frac{1+0}{1+0}$$

$$= \frac{1}{1}$$

$$= 1 \text{ \#.}$$

23) c) $\lim_{n \rightarrow \infty} \frac{7n^2 + 5n + 6}{5n^2 + 4n - 9}$

→ The function takes $\frac{\infty}{\infty}$ form (indeterminate)

$$= \lim_{n \rightarrow \infty} \frac{n^2 \left(7 + \frac{5}{n} + \frac{6}{n^2} \right)}{n^2 \left(5 + \frac{4}{n} - \frac{9}{n^2} \right)}$$

$$= \lim_{n \rightarrow \infty} \frac{7 + \frac{5}{n} + \frac{6}{n^2}}{5 + \frac{4}{n} - \frac{9}{n^2}}$$

$$= \frac{7+0+0}{5+0+0}$$

$$= \frac{7}{5} \text{ \#.}$$

24) d) $\lim_{n \rightarrow \infty} \frac{8n^2 + 7n - 10}{2n^3 - 3n - 4}$

→ The function takes $\frac{\infty}{\infty}$ form (indeterminate)



$$= \lim_{n \rightarrow \infty} \frac{n^2 \left(8 + \frac{7}{n} - \frac{10}{n^2} \right)}{n^3 \left(2 + \frac{3}{n^2} - \frac{4}{n^3} \right)}$$

$$= \frac{8 + \frac{7}{\infty} - \frac{10}{\infty}}{n \left(2 + \frac{3}{\infty} - \frac{4}{\infty} \right)}$$

$$= \frac{8 + 0 - 0}{\infty (2 + 0 - 0)}$$

$$= \frac{8}{\infty}$$

$$= 0$$

25) e) $\lim_{n \rightarrow \infty} \frac{(3n-1)^8 \cdot (4n-1)^2}{(3n+1)^{10}}$

→ The function takes $\frac{\infty}{\infty}$ form (indeterminate)

$$= \lim_{n \rightarrow \infty} \frac{\left\{ n \left(3 - \frac{1}{n} \right) \right\}^8 \cdot \left\{ n \left(4 - \frac{1}{n} \right) \right\}^2}{\left\{ n \left(3 + \frac{1}{n} \right) \right\}^{10}}$$

$$= \lim_{n \rightarrow \infty} \frac{n^8 \left(3 - \frac{1}{n} \right)^8 \cdot n^2 \left(4 - \frac{1}{n} \right)^2}{n^{10} \left(3 + \frac{1}{n} \right)^{10}}$$

$$= \frac{\cancel{27^{10}} \left(3 - \frac{1}{\infty}\right)^8 \cdot \left(4 - \frac{1}{\infty}\right)^2}{\cancel{27^{10}} \left(3 + \frac{1}{\infty}\right)^{10}}$$

$$= \frac{(3-0)^8 \cdot (4-0)^2}{(3+0)^{10}}$$

$$= \frac{3^8 \times 4^2}{3^{10}}$$

$$= \frac{16}{3^{10-8}}$$

$$= \frac{16}{9}$$

26 f) $\lim_{n \rightarrow \infty} (\sqrt{n+a} - \sqrt{n})$

→ The function takes $\infty - \infty$ form
(indeterminate)

$$= \lim_{n \rightarrow \infty} \frac{(\sqrt{n+a} - \sqrt{n}) \times (\sqrt{n+a} + \sqrt{n})}{(\sqrt{n+a} + \sqrt{n})}$$

$$= \lim_{n \rightarrow \infty} \frac{(\sqrt{n+a})^2 - (\sqrt{n})^2}{\sqrt{n+a} + \sqrt{n}}$$

$$= \lim_{n \rightarrow \infty} \frac{\cancel{n} + a - \cancel{n}}{\sqrt{n+a} + \sqrt{n}}$$

$$= \frac{a}{\sqrt{\infty+a} + \sqrt{\infty}}$$

$$= \frac{a}{\infty}$$

$$= 0 \neq.$$

2) $\lim_{x \rightarrow \infty} (\sqrt{x-a} - \sqrt{x-b})$

→ The function takes $\infty - \infty$ form.

$$= \lim_{x \rightarrow \infty} (\sqrt{x-a} - \sqrt{x-b}) \times \frac{(\sqrt{x-a} + \sqrt{x-b})}{(\sqrt{x-a} + \sqrt{x-b})}$$

$$= \lim_{x \rightarrow \infty} \frac{(\sqrt{x-a})^2 - (\sqrt{x-b})^2}{(\sqrt{x-a} + \sqrt{x-b})}$$

$$= \lim_{x \rightarrow \infty} \frac{x-a-x+b}{(\sqrt{x-a} + \sqrt{x-b})}$$

$$= \lim_{x \rightarrow \infty} \frac{b-a}{(\sqrt{x-a} + \sqrt{x-b})}$$

$$= \frac{b-a}{\sqrt{\infty-a} + \sqrt{\infty-b}}$$

$$= \frac{b-a}{\infty}$$

$$= 0 \neq.$$

28 h) $\lim_{n \rightarrow \infty} (\sqrt{3n} - \sqrt{n-5})$

→ The function takes $\infty - \infty$ form.

$$= \lim_{n \rightarrow \infty} (\sqrt{3n} - \sqrt{n-5}) \times \frac{\sqrt{3n} + \sqrt{n-5}}{\sqrt{3n} + \sqrt{n-5}}$$

$$= \lim_{n \rightarrow \infty} \frac{[(\sqrt{3n})^2 - (\sqrt{n-5})^2]}{\sqrt{3n} + \sqrt{n-5}}$$

$$= \lim_{n \rightarrow \infty} \frac{3n - n + 5}{\sqrt{3n} + \sqrt{n-5}}$$

$$= \lim_{n \rightarrow \infty} \frac{2n + 5}{\sqrt{3n} + \sqrt{n}}$$

$$= \lim_{n \rightarrow \infty} \frac{n \left(2 + \frac{5}{n} \right)}{n \sqrt{\frac{3}{n}} + n \sqrt{\frac{1}{n}}}$$

$$= \lim_{n \rightarrow \infty} \frac{n \left(2 + \frac{5}{n} \right)}{n \left(\sqrt{\frac{3}{n}} + \sqrt{\frac{1}{n}} \right)}$$

$$= \frac{2 + \frac{5}{\infty}}{\sqrt{\frac{3}{\infty}} + \sqrt{\frac{1}{\infty}}}$$

$$= \frac{2+0}{0+0}$$

$$= \frac{2}{0}$$

$= \infty$, limit does not exist.

29 i) $\lim_{n \rightarrow \infty} 2n[\sqrt{n^2-1} - n]$

→ The function take infinite (∞) form.

$$= \lim_{n \rightarrow \infty} 2n[\sqrt{n^2-1} - n] \times \frac{\sqrt{n^2-1} + n}{\sqrt{n^2-1} + n}$$

$$= \lim_{n \rightarrow \infty} \frac{2n[(\sqrt{n^2-1})^2 - n^2]}{\sqrt{n^2-1} + n}$$

$$= \lim_{n \rightarrow \infty} \frac{2n[n^2 - 1 - n^2]}{\sqrt{n^2-1} + n}$$

$$= \lim_{n \rightarrow \infty} \frac{-2n}{\sqrt{n^2(1 - \frac{1}{n^2})} + n}$$

$$= \lim_{n \rightarrow \infty} \frac{-2n}{n(\sqrt{1 - \frac{1}{n^2}} + 1)}$$

$$= \frac{-2}{\sqrt{1 - \frac{1}{\infty}} + 1}$$

$$= \frac{-2}{1+1}$$

$$= \frac{-2}{2}$$

$$= -1 \quad \text{Ans.}$$

5.

30 a) If $\lim_{x \rightarrow 1} \frac{x^2-1}{x-1} = \lim_{x \rightarrow a} \frac{x^3-a^3}{x^2-a^2}$ find the value of 'a'.

→ Soln,

$$\text{or, } \lim_{x \rightarrow 1} \frac{(x+1)(x-1)}{(x-1)} = \lim_{x \rightarrow a} \frac{x^3-a^3}{x^2-a^2}$$

$$\text{or, } \lim_{x \rightarrow 1} (x+1) = \lim_{x \rightarrow a} \frac{(x-a)(x^2+ax+a^2)}{(x+a)(x-a)}$$

$$\text{or, } 1+1 = \frac{a^2+a^2+a^2}{a+a}$$

$$\text{or, } 2 = \frac{3a^2}{2a}$$

$$\text{or, } \frac{4}{3} = a$$

$$\therefore a = \frac{4}{3} \quad \text{Ans.}$$

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Q1) b) If $f(x) = \frac{ax+b}{x+1}$, $\lim_{x \rightarrow \infty} f(x) = 2$ and

$\lim_{x \rightarrow \infty} f(x) = 1$, prove that $f(-2) = 0$

→ Soln,

$$\lim_{x \rightarrow \infty} f(x) = 2$$

$$\text{or, } \lim_{x \rightarrow \infty} \frac{ax+b}{x+1} = 2$$

$$\text{or, } \frac{a \cdot 0 + b}{0 + 1} = 2$$

$$\text{or, } b = 2$$
$$\therefore b = 2$$

Again,

$$\lim_{x \rightarrow \infty} f(x) = 1$$

$$\text{or, } \lim_{x \rightarrow \infty} \frac{ax+b}{x+1} = 1$$

$$\text{or, } \lim_{x \rightarrow \infty} \frac{x \left(a + \frac{b}{x} \right)}{x \left(1 + \frac{1}{x} \right)} = 1$$

$$\text{or, } \frac{a + \frac{1}{\infty}}{1 + \frac{1}{\infty}} = 1$$

or, $\frac{1+0}{1+0} = 1$

$\therefore a = 1$

Now,

$$f(n) = \frac{an+b}{n+1}$$

or, $f(n) = \frac{1 \cdot n + 2}{n+1}$

or, $f(n) = \frac{n+2}{n+1}$

or, $f(-2) = \frac{-2+2}{-2+1}$

or, $f(-2) = \frac{0}{-1}$

$f(-2) = 0$

Proved

Exercise 17.2

Evaluate the following limits.

① 1. $\lim_{n \rightarrow 0} \frac{\sin an}{\sin bn}$

→ The function takes $\frac{0}{0}$ form when $n=0$

$$= \lim_{n \rightarrow 0} \frac{\sin an \times an}{an \times \frac{\sin bn}{bn} \times bn}$$

$$= \lim_{n \rightarrow 0} \frac{1 \times an}{1 \times bn}$$

$$= \frac{a}{b}$$

② 2. $\lim_{n \rightarrow 0} \frac{\sin an}{n \cos \beta n}$

→ The function takes $\frac{0}{0}$ form when $n=0$

$$= \lim_{n \rightarrow 0} \frac{\sin an}{\cos \beta n} \times \frac{1}{n}$$

$$= \lim_{n \rightarrow 0} \frac{\sin an \times an}{an \times \frac{\cos \beta n}{\beta n} \times \beta n} \times \frac{1}{n}$$

$$= \lim_{n \rightarrow 0} \frac{1 \times an}{1 \times \beta n} \times \frac{1}{n}$$

$$= \lim_{n \rightarrow 0} \frac{\sin an}{an} \times a \cdot \lim_{n \rightarrow 0} \frac{1}{\cos \beta n}$$

$$= \lim_{x \rightarrow 0} a \cdot \lim_{x \rightarrow 0} \frac{1}{\cos x}$$

$$= a \cdot 1$$

$$= a$$

3. $\lim_{x \rightarrow 0} \frac{\tan 2x - x}{3x - \sin x}$

→ The function takes $\frac{0}{0}$ form when $x=0$

$$= \lim_{x \rightarrow 0} \frac{\tan 2x \times 2x - x}{3x - \frac{\sin x}{x} \times x}$$

$$= \lim_{x \rightarrow 0} \frac{1 \times 2x - x}{3x - 1 \times x}$$

$$= \lim_{x \rightarrow 0} \frac{2x - x}{3x - x}$$

$$= \lim_{x \rightarrow 0} \frac{x}{2x}$$

$$= \frac{1}{2}$$

4. $\lim_{x \rightarrow 0} \frac{\sin ax + \sin bx}{x}$

→ The function takes $\frac{0}{0}$ form when $x=0$

$$= \lim_{x \rightarrow 0} \frac{\sin ax}{x} + \frac{\sin bx}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\sin ax \times a}{ax} + \frac{\sin bx \times b}{bx}$$

$$= 1 \times a + 1 \times b$$

$$= a + b \quad \checkmark$$

5. $\lim_{n \rightarrow 0} \frac{\sin 5n - \sin 3n}{\sin n}$

→ The function takes $\frac{0}{0}$ form when $n=0$

$$= \lim_{n \rightarrow 0} \frac{\frac{\sin 5n}{5n} \times 5n - \frac{\sin 3n}{3n} \times 3n}{\frac{\sin n}{n} \times n}$$

$$= \lim_{n \rightarrow 0} \frac{1 \times 5n - 1 \times 3n}{1 \times n}$$

$$= \lim_{n \rightarrow 0} \frac{5n - 3n}{n}$$

$$= \lim_{n \rightarrow 0} \frac{2n}{n}$$

$$= 2 \quad \checkmark$$

6. $\lim_{n \rightarrow 0} \frac{\cos 6n \cdot \sin 5n}{\sin 6n}$

→ The function takes $\frac{0}{0}$ form when $n=0$

$$= \lim_{n \rightarrow 0} \frac{\sin 5n}{\sin 6n} \cdot \lim_{n \rightarrow 0} \cos 6n$$

$$= \lim_{n \rightarrow 0} \frac{\frac{\sin 5n}{5n} \times 5n}{\frac{\sin 6n}{6n} \times 6n} \cdot \lim_{n \rightarrow 0} \cos 6n$$

$$= \lim_{n \rightarrow 0} \frac{1 \times 5n}{1 \times 6n} \cdot \cos 0$$

$$= \frac{5}{6} \times 1$$

$$= \frac{5}{6} \neq 0$$

7. $\lim_{x \rightarrow a} \frac{\sin(x-a)}{x^2-a^2}$

→ The function takes $\frac{0}{0}$ form when $x=a$

$$= \lim_{x \rightarrow a} \frac{\sin(x-a)}{x^2-a^2}$$

$$= \lim_{(x-a) \rightarrow 0} \frac{\sin(x-a)}{x-a} \times \frac{(x-a)}{(x+a)(x-a)}$$

$$= \lim_{(x-a) \rightarrow 0} 1 \times \frac{1}{x+a}$$

$$= \lim_{x \rightarrow a} \frac{1}{x+a}$$

$$= \frac{1}{a+a}$$

$$= \frac{1}{2a}$$

8. $\lim_{x \rightarrow 0} \frac{\tan \frac{x}{2}}{\sin x}$

→ The function takes $\frac{0}{0}$ form when $x=0$

$$= \lim_{x \rightarrow 0} \frac{\tan \frac{x}{2}}{\frac{x}{2}} \times \frac{x}{2}$$

$$= \lim_{x \rightarrow 0} \frac{\tan \frac{x}{2}}{\frac{x}{2}} \times \frac{x}{2}$$

$$= \lim_{n \rightarrow 0} \frac{1 \times \frac{n}{2}}{5n}$$

$$= \lim_{n \rightarrow 0} \frac{\frac{n}{2} \times \frac{1}{5n}}$$

$$= \frac{1}{10} \neq 1.$$

9. $\lim_{n \rightarrow 0} \frac{1 - \cos 5n}{n^2}$

→ The Function takes $\frac{0}{0}$ form when $n=0$

$$= \lim_{n \rightarrow 0} \frac{1 - \cos 5n}{n^2} \times \frac{1}{1}$$

$$= \lim_{n \rightarrow 0} \left(1 - 1 + 2 \sin^2 \frac{5n}{2} \right) \times \frac{1}{n^2}$$

$$= \lim_{n \rightarrow 0} 2 \cdot \left(\frac{\sin \frac{5n}{2}}{\frac{5n}{2}} \right)^2 \times \left(\frac{5n}{2} \right)^2 \times \frac{1}{n^2}$$

$$= \lim_{n \rightarrow 0} \cancel{2} \times 1^2 \times \frac{25 \cancel{n^2}}{2 \times \cancel{n^2}} \times \frac{1}{\cancel{n^2}}$$

$$= \frac{25}{2} \neq 1.$$

10. $\lim_{n \rightarrow 0} \frac{1 - \cos pn}{1 - \cos qn}$

→ The function takes $\frac{0}{0}$ form when $n=0$

$$= \lim_{n \rightarrow 0} \frac{1 - 1 + 2 \sin^2 \frac{pn}{2}}{1 - 1 + 2 \sin^2 \frac{qn}{2}}$$

$$= \lim_{n \rightarrow 0} \frac{2 \sin^2 \frac{pn}{2}}{2 \sin^2 \frac{qn}{2}}$$

$$= \lim_{n \rightarrow 0} \left(\frac{\left(\frac{\sin \frac{pn}{2}}{\frac{pn}{2}} \right)^2}{\left(\frac{\sin \frac{qn}{2}}{\frac{qn}{2}} \right)^2} \right) \times \left(\frac{pn}{2} \right)^2$$

$$= \lim_{n \rightarrow 0} \frac{1^2 \times \frac{p^2 n^2}{4}}{1^2 \times \frac{q^2 n^2}{4}}$$

$$= \frac{p^2}{q^2} \checkmark$$

11. $\lim_{n \rightarrow 0} (\operatorname{cosec} n - \cot n)$

→ The function takes $\frac{0}{0}$ form when $n=0$

$$= \lim_{n \rightarrow 0} \frac{1}{\sin n} - \frac{\cos n}{\sin n}$$

$$= \lim_{n \rightarrow 0} \frac{1 - \cos n}{\sin n}$$

$$= \lim_{n \rightarrow 0} \frac{1}{\sin n} \times \left(1 - 1 + 2 \sin^2 \frac{n}{2} \right)$$

$$= \lim_{n \rightarrow 0} \frac{1}{\frac{\sin n}{n} \times n} \times 2 \sin^2 \frac{n}{2}$$

$$= \lim_{n \rightarrow 0} \frac{1}{1 \times n} \cdot 2 \cdot \left(\frac{\sin \frac{n}{2}}{\frac{n}{2}} \right)^2 \times \left(\frac{n}{2} \right)^2$$

$$= \lim_{n \rightarrow 0} \frac{2}{n} \times 1^2 \times \frac{n^2}{4}$$

$$= \lim_{n \rightarrow 0} \frac{n}{2}$$

$$= \frac{0}{2}$$

$$= 0$$

② 12. $\lim_{n \rightarrow \frac{\pi}{2}} (\sec n - \tan n)$

→ The function takes $\infty - \infty$ form when $n = \frac{\pi}{2}$

$$\lim_{n \rightarrow \frac{\pi}{2}} \left(\frac{1}{\cos n} - \frac{\sin n}{\cos n} \right)$$

$$= \lim_{n \rightarrow \frac{\pi}{2}} \left(\frac{1 - \sin n}{\cos n} \right)$$

$$= \lim_{n \rightarrow \frac{\pi}{2}} \frac{\sec n - \tan n \times (\sec n + \tan n)}{(\sec n + \tan n)}$$

$$= \lim_{n \rightarrow \frac{\pi}{2}} \frac{\sec^2 n - \tan^2 n}{\sec n + \tan n}$$

$$= \lim_{n \rightarrow \frac{\pi}{2}} \frac{1}{\sec n + \tan n} \quad (\because \sec^2 n - \tan^2 n = 1)$$

$$= \lim_{n \rightarrow \frac{\pi}{2}} \frac{1}{\frac{1}{\cos n} + \frac{\sin n}{\cos n}}$$

$$= \lim_{n \rightarrow \frac{\pi}{2}} \frac{1}{1} \times \frac{\cos n}{1 + \sin n}$$

$$= \frac{\cos \frac{\pi}{2}}{1 + \sin \frac{\pi}{2}}$$

$$= \frac{0}{1 + 1}$$

$$= 0$$



13. $\lim_{n \rightarrow 0} \frac{\operatorname{cosec} n - \cot n}{n}$

→ The function takes $\frac{0}{0}$ form when $n=0$

$$= \lim_{n \rightarrow 0} \frac{1}{\sin n} - \frac{\cos n}{\sin n}$$

$$= \lim_{n \rightarrow 0} \frac{1 - \cos n}{\sin n} \times \frac{1}{n}$$

$$= \lim_{n \rightarrow 0} \frac{1 - 1 + 2\sin^2 \frac{n}{2}}{\sin n} \times \frac{1}{n}$$

$$= \lim_{n \rightarrow 0} \frac{\frac{1}{n} \times 2 \times \left(\frac{\sin \frac{n}{2}}{\frac{n}{2}} \right)^2 \times \left(\frac{n}{2} \right)^2}{\sin n \times n}$$

$$= \lim_{n \rightarrow 0} \frac{\frac{2}{n} \times 1^2 \times \frac{n^2}{4}}{1 \cdot n}$$

$$= \lim_{n \rightarrow 0} \frac{\frac{n}{2} \times 1}{n}$$

$$= \frac{1}{2}$$

u.

14. $\lim_{n \rightarrow 0} \frac{\tan n^\circ}{n}$

→ The function takes $\frac{0}{0}$ form when $n=0$

$$= \lim_{n \rightarrow 0} \frac{\tan \left(n \times \frac{\pi}{180} \right)}{n}$$

$$= \lim_{x \rightarrow 0} \frac{\tan \frac{\pi}{180} x}{x \times \frac{\pi}{180}} \times \frac{\pi}{180}$$

$$= \lim_{x \rightarrow 0} \frac{1 \times \frac{\pi}{180}}{\frac{\pi}{180}}$$

$$= \frac{\pi}{180} \text{ Ans.}$$

15. $\lim_{x \rightarrow 0} \frac{\sin x}{x}$

→ The function takes $\frac{0}{0}$ form when $x=0$

$$= \lim_{x \rightarrow 0} \frac{\sin \left(x \times \frac{\pi}{200} \right)}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\sin \left(x \times \frac{\pi}{200} \right)}{x \times \frac{\pi}{200}} \times \frac{\pi}{200}$$

$$= \lim_{x \rightarrow 0} \frac{1 \times \frac{\pi}{200}}{\frac{\pi}{200}}$$

$$= \frac{\pi}{200} \text{ Ans.}$$

16. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\tan x + \cot x}{\tan x - \cot x}$

→ The function takes $\frac{\infty}{\infty}$ form when $x = \frac{\pi}{2}$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\tan x + \cot x}{\tan x - \cot x}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x + \cos x}{\cos x \sin x}$$

$$\frac{\sin x - \cos x}{\cos x \sin x}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin^2 x + \cos^2 x}{\sin x \cos x}$$

$$\frac{\sin^2 x - \cos^2 x}{\sin x \cos x}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{1}{\sin^2 x - \cos^2 x}$$

$$= \frac{1}{\sin^2 \frac{\pi}{2} - \cos^2 \frac{\pi}{2}}$$

$$= \frac{1}{1 - 0}$$

$$= \frac{1}{1}$$

$$= 1 \text{ h.}$$

17. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cot x}{\frac{\pi}{2} - x}$

→ The function takes $\frac{0}{0}$ form, when $x = \frac{\pi}{2}$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{\sin x}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin(\frac{\pi}{2} - x)}{(\frac{\pi}{2} - x)} \times \frac{1}{\sin x}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{1 \times 1}{\sin x}$$

$$= \frac{1}{\sin \frac{\pi}{2}}$$

$$= \frac{1}{1}$$

$$= 1 \text{ \#}$$

18. $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\cos x - \sin x}{x - \frac{\pi}{4}}$

→ The function takes $\frac{0}{0}$ form, when $x = \frac{\pi}{4}$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{\cos x}{x - \frac{\pi}{4}} - \frac{\sin x}{x - \frac{\pi}{4}}$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{-\cos x}{\frac{\pi}{4} - x} + \frac{\sin x}{\frac{\pi}{4} - x}$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x}{\frac{\pi}{4} - x} - \frac{\sin \cos x}{\frac{\pi}{4} - x}$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin \left(\frac{\pi}{2} - x \right) - \sin x}{x - \frac{\pi}{4}}$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{2 \cos \left(\frac{\frac{\pi}{2} - x + x}{2} \right) \cdot \sin \left(\frac{\frac{\pi}{2} - x - x}{2} \right)}{x - \frac{\pi}{4}}$$